Technical Guide: Titan T900 Heavy Duty Gas Turbine

Manual Code: TG-T900-OM-2026 | Revision: 8.0

1. Turbine Baseline Thermodynamic Parameters

The Titan T900 is an industrial aeroderivative gas turbine designed for simple-cycle and combined-cycle power generation. The 17-stage axial-flow compressor operates with variable stator vanes (VSV) on the first four stages to prevent aerodynamic stall during cold startup conditions.

Parameter	ISO Baseline Value	Operational Limit / Unit
ISO Base Power Output	45.0 MW	At 15°C ambient, sea level
Thermal Efficiency (Simple Cycle)	38.5%	Lower heating value basis
Heat Rate (LHV)	9,350 kJ / kWh	± 1.5% tolerance
Compressor Pressure Ratio	21.5 : 1	Nominal full-load pressure ratio
Turbine Rotor Speed	3,000 RPM (50Hz) / 3,600 RPM (60Hz)	Driven via epicyclic reduction gearbox
Exhaust Gas Mass Flow	138.5 kg / s	At base load rating
Exhaust Gas Temperature	540.0°C (1,004°F)	Exhaust diffuser plenum exit
NOx Emissions (Dry Low NOx)	< 15 ppmvd	Corrected to 15% O2 on natural gas

2. Fuel Delivery & Conditioning Systems

The T900 combustion system incorporates 14 dry low emission (DLE) combustor cans arranged in a can-annular configuration.

2.1 Fuel Gas Quality & Filtration Requirements

Natural gas supplied to the turbine skid must arrive at a regulated pressure of 28.0 to 32.0 bar gauge (barG). Gas temperature at the fuel control valve inlet must maintain a minimum of 20°C superheat above hydrocarbon and water dew points. The primary duplex fuel coalescing filters must provide 3.0 microns absolute filtration rating. Total sulfur content must not exceed 25 ppm weight.

2.2 Dual-Fuel Transfer Procedure

When transferring from fuel gas to backup distillate diesel (Ultra Low Sulfur Diesel, ASTM D975), the automated system opens the liquid purge valve and engages the high-pressure fuel pump at 45% load. Transfer duration must complete within 120 seconds to prevent thermal shock to the first-stage nozzle guide vanes.

3. Lubrication Oil System & Bearing Temperatures

The turbine lube oil skid provides pressurized lubrication and hydraulic control fluid to the journal bearings and reduction gearbox.

3.1 Lube Oil Specifications

The required lubricant is ISO VG 46 synthetic ester turbine oil conforming to GEK-32568 and MIL-PRF-23699 specifications. The main stainless steel oil reservoir has an active capacity of 4,500 liters. The normal lube oil supply temperature to the bearings must be maintained between 45.0°C and 50.0°C by the thermostatically controlled plate heat exchanger.

3.2 Bearing Temperature Trip Limits

Each hydrodynamic tilt-pad journal bearing is equipped with dual duplex PT100 RTDs embedded in the babbitt metal layer.
- Normal Operating Temperature: 65°C to 75°C
- High Temperature Alarm: 85.0°C
- Emergency Turbine Trip Limit: 105.0°C

If lube oil header pressure drops below 1.2 bar gauge, the DC emergency auxiliary pump engages within 0.8 seconds.

4. Rotor Coupling Torquing & Mechanical Overhaul

Precision mechanical assembly is critical during major inspection outages (hot gas path inspection every 24,000 equivalent operating hours).

4.1 Rotor Coupling Bolt Preload Specifications

The main rotor flange coupling connects the turbine shaft to the load gear pinion via 16 high-tensile nickel alloy bolts (M36 x 3.0, Grade 10.9). All bolts must be torqued using a calibrated hydraulic torque wrench in a diagonal criss-cross star sequence in three distinct stages:
- Stage 1: Snug torque to 350 Nm
- Stage 2: Intermediate torque to 800 Nm
- Stage 3: Final certified torque to 1,450 Nm (± 25 Nm)
CRITICAL: Apply MoS2 molybdenum disulfide anti-seize paste to threads and washer contact surfaces prior to torque sequence.

5. Emergency Trip Setpoints & Safety Interlocks

The safety management system (SMS) provides independent SIL 3 emergency trip logic across key physical parameters.

Monitored Parameter	Sensor Instrument	Warning Alarm Setpoint	Emergency Trip Setpoint
Rotor Radial Vibration	Bently Nevada 3300 Proximity Probes	4.5 mm / s RMS	7.1 mm / s RMS (Instantaneous Trip)
Turbine Shaft Overspeed	Magnetic Speed Pickups (3 redundant)	3,150 RPM (105%)	3,300 RPM (110% of rated speed)
Lube Oil Header Pressure	Dual Differential Pressure Transmitters	1.5 barG	1.0 barG (Trip in 1.5 seconds)
Exhaust Temperature Spread	24 Chromel-Alumel Thermocouples	Spread > 35°C	Spread > 50°C (Combustor Flameout Trip)
Compressor Inlet Differential Pressure	Delta-P Manometer	8.0 mbar	12.0 mbar (Filter collapse warning)

5.1 Emergency Cooldown Barring Gear Cycle

Following any emergency trip from base load, the automatic hydraulic barring gear (turning gear) must rotate the shaft at 45 RPM continuously for 48 hours. Turning prevents asymmetric thermal rotor sag (thermal bow), which can cause catastrophic blading rubbing.